The Journal of Open Source Software

NGSBEM: potential operators for FEM-BEM couplings in NETGEN/NGSolve

Rafael Dorigo [1*], **Joachim Schöberl** [1*], **and Lucy Weggler** [2*¶]

1 TU Wien, Austria **2** HTW Berlin, Germany ¶ Corresponding author * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 03 December 2025
Published: unpublished

Summary

NGSBEM is the dedicated boundary element method (BEM) module within the NETGEN/NGSolve framework (NGSolve developers, 2026c). It provides the full set of tools needed to formulate and solve boundary integral equations, including potential operators and their Galerkin discretizations. Seamlessly integrated with the NGSolve core, NGSBEM enables efficient numerical solutions of linear interior and exterior boundary value problems based on the BEM.

Key features of NGSBEM, in conjunction with NGSolve, include:

- Use of finite element spaces already available in NGSolve, including high-order elements on curved surfaces.
- A user-friendly Python interface.
- Declarative programming for problem formulation and solution.
- Standard potential operators for electrostatics, acoustics, and electromagnetics, allowing both scalar and vector-valued densities to be discretized.
- Support for multi-domain problems through FEM–BEM coupling.
- Acceleration via a Multi-Level Fast Multipole Method (MLFMM) for approximating the arising Galerkin matrices.
- Kernel-driven implementations of potential operators, enabling users to add custom operators.
- Extensive documentation and demonstrations supporting users as they start working with.

The source code is publicly available at (NGSolve developers, 2026a).

Statement of Need

Computational engineering frequently involves solving boundary value problems (BVPs), including linear and nonlinear partial differential equations, as well as interior, exterior, and transmission problems. While the finite element method (FEM) excels at handling interior problems with nonlinearities, the BEM is the method of choice for solving linear second-order problems in unbounded domains.

A unified framework that naturally combines both approaches is rare. NETGEN/NGSolve addresses this gap by providing a complete environment for Galerkin BEM and its coupling with FEM.

State of the Field

A number of open-source software packages for FEM, BEM, and FEM-BEM coupling are available, but fully integrated, high-order frameworks combining both methods in a single environment remain limited. On the FEM side, widely used libraries such as FEniCS (FEniCS developers, 2026) and deal.II (deal.II developers, 2026) provide advanced high-order discretizations, including support for quadrilateral/hexahedral elements and curved geometries, but do not natively include BEM functionality. Conversely, dedicated BEM libraries such as Bempp (Bem++ developers, 2026) offer a mature operator-based framework with documented FEM-BEM coupling interfaces (e.g., to FEniCS), yet are generally restricted to piecewise planar triangular surface meshes and do not provide a unified high-order curved-element infrastructure.

Hybrid approaches exist but are typically loosely coupled. For example, FreeFEM (FreeFEM developers, 2026) supports BEM formulations through external libraries such as BEMTool, but focuses primarily on simplex-based discretizations with moderate polynomial order. Other frameworks, such as Bempp, follow a similar paradigm of specialized BEM components coupled to FEM solvers rather than a unified framework.

In summary, alternatives do exist, but they are not directly comparable in terms of integration: most either provide high-order FEM without BEM, or BEM with limited geometry/order support and external coupling. Fully unified environments that combine high-order FEM, BEM, curved elements, and native coupling - as realized in NETGEN/NGSolve - remain comparatively rare. This justifies a dedicated implementation rather than extending existing loosely coupled frameworks.

Software Design

The architecture of NGSolve (NGSolve developers, 2026a), developed in conjunction with Netgen and extended by NGSBEM, reflects a deliberate balance between usability, extensibility, and high-performance execution, aligned with the needs of computational research. A key design decision is the separation between a high-level Python interface and a performance-critical C++ core. This allows users to express FEM-BEM couplings and variational formulations concisely, while delegating assembly, discretization, and solver execution to optimized backend components.

The core leverages template-based generic programming to achieve compile-time specialization of finite element operators, enabling efficient handling of high-order and curved elements without sacrificing generality. Parallelism is addressed through MPI and shared-memory threading, ensuring scalability for large-scale simulations. The trade-off is increased implementation complexity, which is mitigated by a consistent decomposition into composable abstractions (finite element spaces, bilinear forms, integrators, and boundary operators).

Importantly, the architecture is designed for incremental extensibility: new methods can be introduced at the level of individual components without modifying the overall system. This has enabled a contribution-driven evolution, where features such as FEM-BEM coupling originated as external developments before integration into the core. Such a design directly supports reproducible research by allowing rapid prototyping while maintaining a stable, performant foundation.

The documentation is publicly available at (NGSolve developers, 2026b).

Research impact statement

The NGSBEM extension within NGSolve has already demonstrated early but tangible research impact, particularly through increasing adoption and engagement within the existing NGSolve user community. While the software is relatively recent, there is clear evidence of growing external interest and active use in emerging research problems. For example, initial applications in low-frequency stabilization have led to publicly shared demonstration cases and triggered follow-up on robust preconditioners for Maxwell-type problems (Andriulli et al., 2021) which are now part of the showcased examples on (NGSBem developers, 2026).

In addition, recent user interactions - particularly through developer and user meetings - indicate a rising demand for FEM-BEM coupling capabilities, with multiple groups exploring how NGSBEM can be applied to new classes of problems. This organic uptake reflects both the relevance of the approach and the accessibility of the implementation. The project benefits from the established ecosystem of NGSolve, including documented examples, reproducible scripts, and a stable user base, which enable external validation and reuse. This lowers the barrier for adoption in independent research settings.

Although formal publications and large-scale benchmarks are still emerging, the current trajectory provides credible evidence of near-term impact. The present publication represents the first consolidated description of NGSBEM and is expected to further enhance visibility, reproducibility, and adoption within the computational science community. Early adoption within the existing NGSolve community provides a concrete indicator of practical relevance.

Usability, Accuracy, and Efficiency

The BEM is attractive due to its dimensional reduction and high accuracy. However, it also introduces challenges, such as singular integral operators and dense matrices. NGSBEM is designed to resolve these difficulties internally, providing robust and efficient implementations while sparing users from the underlying complexity.

Usability

NGSolve follows the philosophy of allowing users to "type formulas as they appear in the book." NGSBEM extends this philosophy to the BEM: users express boundary integral operators declaratively in Python, in a syntax that mirrors their mathematical formulation. This clarity and directness in representing operators is a central design goal.

Consider, for example, the Dirichlet problem

$$\begin{cases} \Delta\phi = 0, & \text{in } \Omega, \\ \gamma_0\phi = m, & \text{on } \Gamma. \end{cases}$$

The representation formula yields:

$$\phi(\mathbf{x}) = \mathrm{LaplaceSL}(j)(\mathbf{x}) - \mathrm{LaplaceDL}(m)(\mathbf{x})\,. \tag{1}$$

To compute the unknown Neumann data j, one solves the Galerkin discretization of the single layer operator. With trial functions u_j and test functions v_i, the matrix entries are

$$V_{ij} = \int_\Gamma \left[\mathrm{LaplaceSL}(u_j)\right] v_i\, ds.$$

In NGSBEM, the matrix is assembled as:

```
fesL2 = SurfaceL2(mesh, order=3, dual_mapping=True)
u, v = fesL2.TnT()
V = LaplaceSL(u * ds) * v * ds
```

Given V, the linear system

$$Vj = b$$

is solved using a standard iterative solver.

A second example illustrates the hypersingular operator arising for the Neumann problem. Its Galerkin matrix is given by

$$D_{ij} = \int_\Gamma \left[\mathrm{LaplaceSL}(\mathbf{curl}_\Gamma u_j)\right] \mathbf{curl}_\Gamma v_i(\boldsymbol{x})\, \mathrm{d}s_x \, .$$

implemented simply as

```
D = LaplaceSL(curl(u) * ds) * curl(v) * ds
```

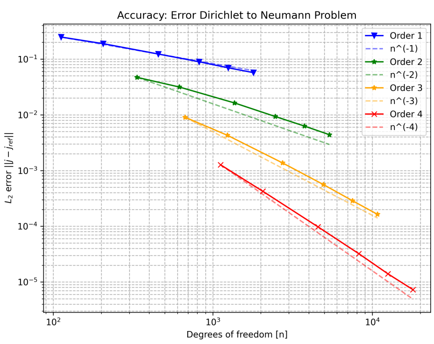
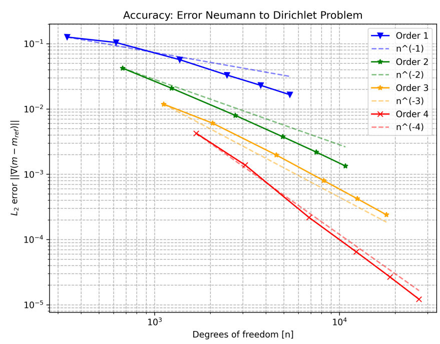
The linear system with Galerkin matrix D is solved using a standard iterative solver.

Single and double layer operators for the Helmholtz and Maxwell equations are provided analogously.

Accuracy

Accurate integration is central to the BEM, particularly because singular kernels must be treated with care. NGSBEM employs the specialized quadrature rules of Sauter and Schwab (Sauter & Schwab, 2011), which support curved elements and enable high-order convergence.

The numerical experiments below compare the computed Cauchy data with the known exact solution for the Laplace equation. The results demonstrate optimal convergence rates.

Laplace Dirichlet Problem	Laplace Neumann Problem
	

Beyond scalar-valued problems, NGSBEM supports vector finite elements and BEM operators for electromagnetics, with numerical evidence of optimal convergence documented in (**?**).

Another challenge is the accurate evaluation of potentials, especially in the near field. NGSBEM implements techniques developed by Gumerov and others (Nail A. Gumerov & Duraiswami, 2021; Kaneko et al., 2023). Depending on the geometric configuration, triangles are either subdivided or a line-based approximation is applied, ensuring stable and accurate potential evaluation.

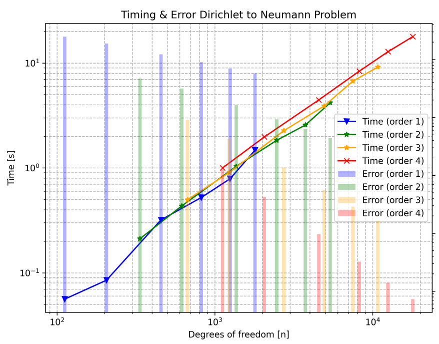
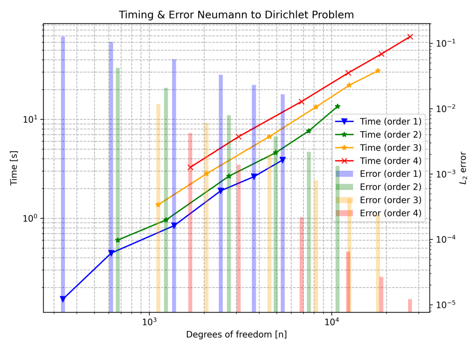
Efficiency

Galerkin matrices are generated on the fly using the multi-level fast multipole method (MLFMM) of Greengard and Rokhlin (Greengard & Rokhlin, 1987), with efficient computation of translation and rotation coefficients following the work of Gumerov and Duraiswami (Dorigo, 2025; Nail A. Gumerov & Duraiswami, 2001; Nail A. Gumerov & Duraiswami, 2015).

Performance is further enhanced by:

- the use of AVX vectorization,
- full parallelization across all major computational stages.

To assess efficiency, we compare accuracy and runtime across different polynomial orders. Higher-order elements provide significantly better accuracy at comparable runtime, despite the increased number of unknowns.

Laplace Dirichlet Problem	Laplace Neumann Problem
	

For example, in Laplace Dirichlet Problem with efficiency plot shown on the left:

- order 1 at 1250 DOFs (0.789 s runtime, error 0.07), vs.,
- order 4 at 1120 DOFs (1.000 s runtime, error 0.001)

This corresponds to an error reduction of approximately 98.6% at comparable runtime.

Similar improvements appear across other configurations, demonstrating the practical efficiency of higher-order BEM.

All numerical experiments were run on a MacBook Pro equipped with an Apple M4 Pro (14 cores: 10 performance + 4 efficiency) and 48 GB of RAM. Timings were measured on the host machine under macOS.

Example

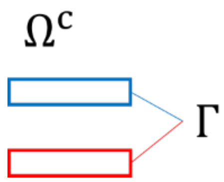
Solving a PDE with the BEM proceeds in two steps:

1. solve an integral equation for the unknown boundary trace(s),
2. evaluate the representation formula to obtain the solution in the domain.

Consider a plate capacitor:

The Journal of Open Source Software

$$
\begin{aligned}
-\Delta\phi &= 0, && \text{in } \Omega^c\,,\\
\gamma_0\phi &= m\,, && \text{on } \Gamma\,,\\
\lim_{\|x\|\to\infty} \phi(x) &= \mathcal{O}\left(\frac{1}{\|x\|}\right)\,, && \|x\|\to\infty\,.
\end{aligned}
$$



Using Equation 1, the unknown Neumann data j is computed via a BEM system:

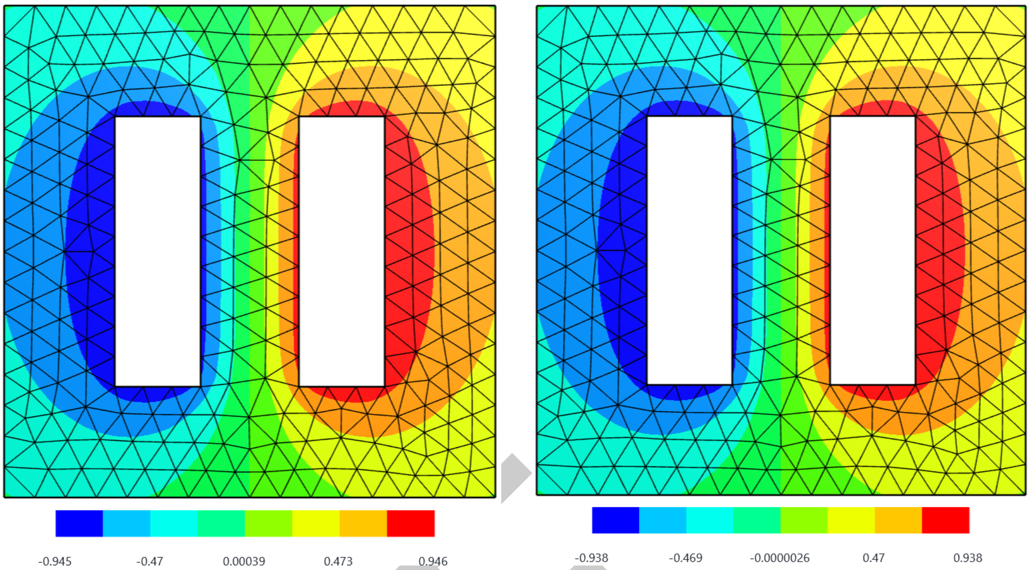
```
j = GridFunction(fesL2)

# generate potential operators V, M and K
with TaskManager():
    SLPotential = LaplaceSL(u*ds(bonus_intorder=3))
    DLPotential = LaplaceDL(uH1*ds(bonus_intorder=3))
    V = SLPotential*v*ds(bonus_intorder=3)
    K = DLPotential*v*ds(bonus_intorder=3)
    M = BilinearForm(uH1*v*ds(bonus_intorder=3)).Assemble()

# solve the linear system V j = (-1/2 M + K) m
with TaskManager():
    pre = BilinearForm(u.Trace()*v.Trace()*ds, diagonal=True).Assemble().mat.Inverse()
    rhs = ((-0.5 * M.mat + K.mat) * m).Evaluate()
    CG(
      mat = V.mat, pre=pre, rhs = rhs, sol=j.vec, tol=1e-8, maxsteps=200,
      initialize=False, printrates=False
      )
```

Afterwards, the potential ϕ can be evaluated anywhere. For example, on a screen cutting through the plates:

```
screen = WorkPlane(
  Axes((0,0,0), X, Z)).RectangleC(4, 4).Face()
  - Box((-1.1,-1.1,0.4), (1.1,1.1,1.1))
  - Box((-1.1,-1.1,-1.1), (1.1,1.1,-0.4)
  )
mesh_screen = Mesh(OCCGeometry(screen).GenerateMesh(maxh=0.25)).Curve(1)
fes_screen = H1(mesh_screen, order=3)
phi_on_screen = GridFunction(fes_screen)
print ("ndofscreen=", fes_screen.ndof)
with TaskManager():
    phi_on_screen.Set(
      -SLPotential(j)+DLPotential(utop)+DLPotential(ubot),
      definedon=mesh_screen.Boundaries(".*"), dual=False
      )
Draw (gf_screen)
```

The BEM solution (left) and a FEM reference solution (right) are shown here:

BEM Solution ϕ	FEM Solution ϕ_{fem}



Because the problem is exterior, the FEM requires either infinite elements or an artificial boundary with prescribed data. The BEM, in contrast, handles the unbounded domain naturally and without mesh refinement at edges or corners. The maximum discrepancy between FEM and BEM is around 2.5%, occurring near corner singularities.

Further examples, including FEM–BEM coupling, mixed formulations, and acoustic or electromagnetic scattering, can be found in the online documentation (NGSBem developers, 2026).

Details on the theoretical background for high order BEM for electrostatics and electromagnetics, can be found in (Weggler, 2011, 2012).

AI usage disclosure

No AI-assisted tools were used in the development of the software described in this work. The codebase of NGSolve and its extensions, including NGSBEM, has been developed through conventional scientific programming practices by domain experts over multiple decades. AI tools were also not used in the generation of the scientific results presented in this manuscript.

References

Andriulli, F. P., Cools, K., Bağcı, H., & Zutter, D. D. (2021). Electromagnetic integral equations: Insights in conditioning and preconditioning. *IEEE Open Journal of Antennas and Propagation*, *2*, 1234–1247. https://doi.org/10.1109/OJAP.2021.3121097

Bem++ developers. (2026). *Bempp-cl website*. https://bempp.com

deal.II developers. (2026). *Deal.II website*. https://www.dealii.org

Dorigo, R. (2025). *High order boundary element methods accelerated by the fast multipole method in NGSolve* [Diploma Thesis, Technische Universität Wien]. https://doi.org/10.34726/hss.2025.136548

FEniCS developers. (2026). *FEniCS website*. https://fenicsproject.org

FreeFEM developers. (2026). *FreeFEM website*. https://freefem.org

Greengard, L., & Rokhlin, V. (1987). A fast algorithm for particle simulations. *Journal of Computational Physics*, *73*(2), 325–348. https://doi.org/10.1016/0021-9991(87)90140-9

Gumerov, Nail A., & Duraiswami, R. (2001). *Fast, exact, and stable computation of multipole translation and rotation coefficients for the 3-d helmholtz equation*. https://drum.lib.umd.edu/items/f0c54406-2abe-4416-b35f-1d4f669b8c50

Gumerov, Nail A., & Duraiswami, R. (2015). Recursive computation of spherical harmonic rotation coefficients of large degree. In R. Balan, M. J. Begué, J. J. Benedetto, W. Czaja, & K. A. Okoudjou (Eds.), *Excursions in harmonic analysis, volume 3: The february fourier talks at the norbert wiener center* (pp. 105–141). Springer International Publishing. https://doi.org/10.1007/978-3-319-13230-3_5

Gumerov, Nail A., & Duraiswami, R. (2021). *Analytical computation of boundary integrals for the helmholtz equation in three dimensions*. https://arxiv.org/abs/2103.17196

Kaneko, S., Gumerov, N. A., & Duraiswami, R. (2023). *Recursive analytical quadrature of laplace and helmholtz layer potentials in* $\mathbb{R}^3$. https://arxiv.org/abs/2302.02196

NGSBem developers. (2026). *NGSBem repro with tutorials and convergence tests*. https://weggler.github.io/docu-ngsbem/intro.html

NGSolve developers. (2026a). *NGSolve*. https://github.com/NGSolve/ngsolve

NGSolve developers. (2026b). *NGSolve tutorials*. https://docu.ngsolve.org

NGSolve developers. (2026c). *NGSolve website*. https://ngsolve.org

Sauter, S., & Schwab, C. (2011). *Boundary element methods* (Vol. 39). Springer. https://doi.org/10.1007/978-3-540-68093-2

Weggler, L. (2011). *High order boundary element methods* [Dissertation]. Saarland University. https://publikationen.sulb.uni-saarland.de/bitstream/20.500.11880/26312/1/thesis_weggler_final_6.1.12.pdf

Weggler, L. (2012). Stabilized boundary element methods for low-frequency electromagnetic scattering. *Mathematical Methods in the Applied Sciences*, *35*(5), 574–597. https://doi.org/10.1002/mma.1597